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Preparation of Polypropylene Graft Maleic Anhydride (PP-g-MA) via Twin Screw Extrusion

Rittirong Pruthitikul^{1, a} and Pitcha Liewchirakorn^{1, b}

^{1,2}Plastics Technology Laboratory, National Metal and Materials Technology Center,
National Science and Technology Development Agency,
Pathumthani, 12120 Thailand

^arittirop@mtec.or.th, ^bpitchal@mtec.or.th

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Abstract. Polypropylene (PP) samples grafted with Maleic Anhydride (MA) were prepared by reactive blending via co-rotating twin screw extrusion. The PP was functionalized in the presence of an optimized amount of precursor, Dicumyl Peroxide (DCP) and MA. The amounts of MA grafted on PP were checked by standard test method for acid number. Maleated PP as a compatibilizer with the highest acid number was added to PP/Polyamide 6 (PA6) and blended in the twin screw extrusion. It was found that the PP-g-MA enhanced tensile properties as well as the izod impact properties of notched samples of PP/PA6/PP-g-MA blends compared to neat polypropylene. The scanning electron microscope confirmed the good adhesion of PA6 on PP matrix.

Introduction

Commodity plastics show inferior properties such as mechanical properties, heat resistance, and barrier properties compared to engineering plastics [1-3]. Yet, the low price makes them more attractive for present belt-tightening economy. Blending of two or more polymers proposes an interesting route to modifying the properties of plastics. It is a cost-effective way to produce new materials with desired sets of combined properties [1-4]. In this work, we prepared PP-g-MA via twin screw extrusion. Free radical grafting reactions of MA onto PP using Dicumyl Peroxide (DCP) as the initiator were investigated in detail to study the grafting efficiency and optimize the grafting conditions. Effects of the DCP concentration on the degree of grafting were also investigated. The maleated PP was then used as compatibilizer to promote better adhesion between PP and PA6. Its effectiveness as a compatibilizer was compared with a commercial product Polybond.

Methodology

Materials. Dicumyl peroxide (AR grade), Phenolphthalein indicator (AR grade), Potassium hydroxide (AR grade), and Maleic anhydride (GR grade) were supplied by Fluka. Organic solvents (AR grade); xylene, methanol, and ethanol were supplied by LABSCAN. Polypropylene was supplied by HMC Polymers Company Limited (PW-888N MFI 60 g/10 min and HP480S MFI 40 g/10 min). PA6 (1013B) was supplied by UBE Group (Thailand). Polybond, a commercial polypropylene grafted maleic anhydride was produced by Crompton Polybond, Co. All the reagents were used as received.

Reactive Blends Preparation. Reactive blends of different compositions were prepared using a JSW twin screw extruder model TEX30 α (42:1 L/D Ratio), at screw rotation speed of 100 rpm. Its 12 heating zones were set at temperature ranging between 180-210 °C. The blends consisted of 94 to 95.5 wt% of PP (HP480S), 4 wt% of MA and 0.5 to 2 wt% of DCP. The blend compositions are presented in Table 1.

Table 1 Mass ratio of PP/MA/DCP

	PP (HP480S)	MA	DCP
Formula 1	95.5	4	0.5
Formula 2	95	4	1
Formula 3	94.5	4	1.5
Formula 4	94	4	2

Evaluation of Acid Number. In this test method used for determining the acid number in PP-g-MA compatibilizer, the measured numbers were obtained by direct titration of material indicating the amount of free acid. 10 g of PP-g-MA compatibilizer was transferred to 500 ml. beaker. 350 ml. of xylene was then added. The content was heated and stirred at 140 °C until completely melt. The melt was filled and stirred spontaneously into 700 ml. methanol to promote precipitation. The precipitated solid was percolated and kept in a 60 °C oven for 24 h. Following the drying process, 1 g of dry white lees, weighed to the nearest 0.0001 g, was transferred to a 250 ml. flask with 40 ml. of xylene. Samples were heated on a hot plate to put into solutions. Occasional swirl may be necessary. 3-5 drops of Phenolphthalein indicator, 1% in ethanol was added and titrated the hot solution to the first persistent pink color. The end point was taken when the pink color remained for at least 10 s.

The acid numbers were calculated as follows:

Acid number (mg KOH/g) = (AN x 56.1)/grams of sample used.

Where: A = milliliters of KOH solution required for titration of the sample,

N = normality of the alkali solution

PP/PA6/PP-g-MA Blends Preparation. PP (PW-888N) /PA6/PP-g-MA blend 65/30/5 wt% was prepared using twin screw extrusion. Screw rotation speed was set at 200 rpm, with 12 heating zones at temperature ranging between 190-220°C. Prior to blending, all contents were dried for 12 h at 80 °C under vacuum. Following the blending process, PP/PA6/PP-g-MA blend pellets were shaped to 130x150x3.2 mm³ by compression molding machine with temperature 230°C, Dumbbell specimens were prepared by a die cut tool.

Characterization. The tensile tests for PP/PA6/PP-g-MA blends were conducted according to ASTM D 638-99 with Instron testing machine model 55R4502 at a crosshead speed of 50 mm/min. For impact test, notched specimens were tested by instrument impact tester, Radmana model ITR2000. The failure mode of PP/PA6/PP-g-MA blends was characterized by Scanning Electron Microscopy (SEM) performed on fractured surfaces under liquid nitrogen condition.

Results and Discussion

The acid numbers of maleated PP are shown in Fig. 1 It is clearly seen that acid numbers successively increase with increasing DCP contents from 0.5 to 1.5 wt% in the PP/MA/DCP reactive blends, indicating that DCP can drive the MA grafting reaction. At 2 wt%, DCP becomes excessive resulting in no further increase in acid number. With 4 wt% MA in the system, 1.5 wt% of DCP yield the maximum acid number of 5.61 mg KOH/g, which is higher than the acid number of 5.13 mg KOH/g acquired from the commercial product Polybond.

For PP/PA6/Compatibilizer blends, tensile strength and elongation at break are lower than PA6 but higher than PP due to the effect of PA6 composite reinforcement in the matrix of PP (see Fig. 2 and Fig. 3). Considering impact strength in Fig. 4, the values of impact strength of PP/PA6/Compatibilizer blends are clearly higher than PP due to better energy absorption in PA6 phase and its interface. These benefits could only be obtained from effective compatibilization in blends of PP and PA6 with only 5 wt% of compatibilizer evidently shown in Fig. 5 by small particle size of PA6 dispersed in PP matrix. The fractured surfaces of PP/PA6/compatibilizer blends are shown in Fig 5. The failure mode of these systems is cohesive failure on PP matrix. The good

adhesion between PA6 and PP results from maleated PP compatibilizer both in the case of laboratory synthesized PP-g-MA (Fig. 5a) and Polybond (Fig. 5b).

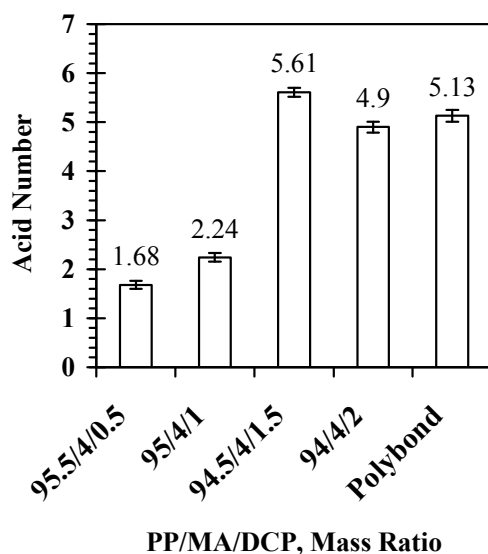


Fig. 1 Effect of precursor compositions on acid numbers

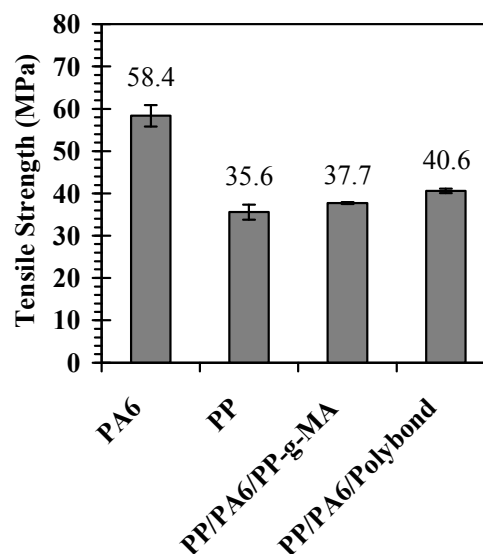


Fig. 2 Tensile strength of PA6, PP, and PP/PA6/Compatibilizer blends

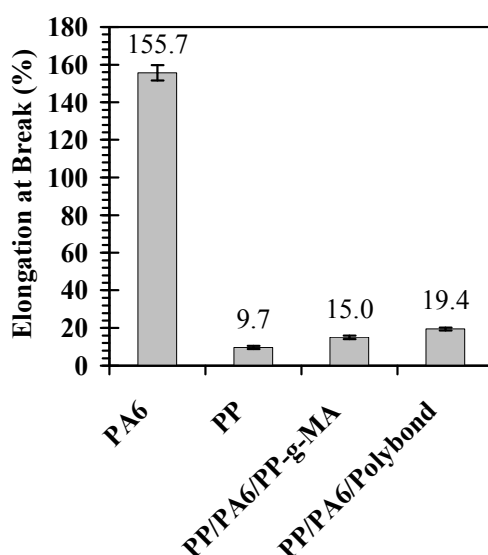


Fig. 3 Elongation at break of PA6, PP, and PP/PA6/Compatibilizer blends

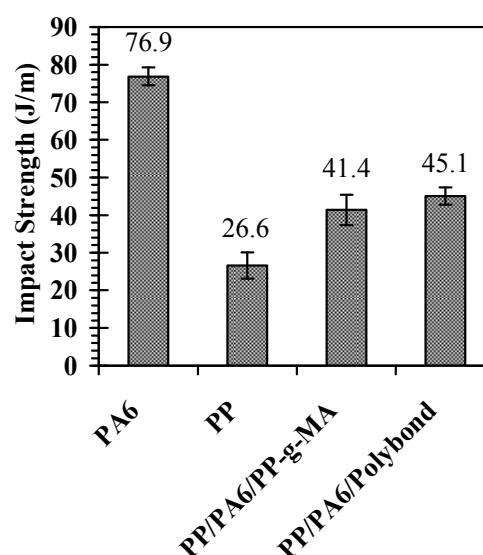


Fig. 4 Izod impact strength of PA6, PP, and PP/PA6/Compatibilizer blends

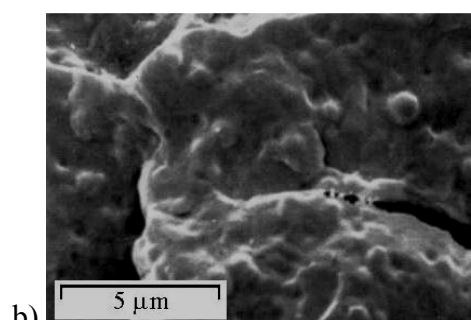
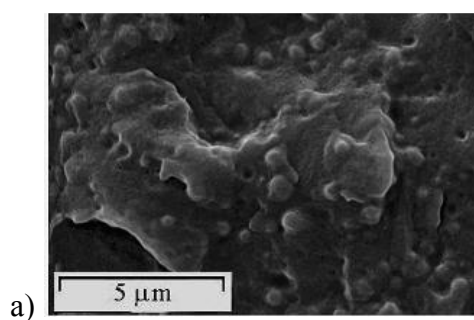


Fig. 5 Morphology of PP/PA6/compatibilizer blends a) PP/PA6/PP-g-MA blend and b) PP/PA6/Polybond blend

In comparing the effectiveness between our PP-g-MA and Polybond, it can be seen that PP/PA6/Polybond has a slightly better performance compared to our PP/PA6/PP-g-MA in term of strength, (40.6 MPa with Polybond compatibilizer, 37.7 MPa with our PP-g-MA compatibilizer) elongation at break, (19.4 % with Polybond compatibilizer, 15 % with our PP-g-MA compatibilizer) and impact strength (45.1 J/m with Polybond compatibilizer, 41.4 J/m with our PP-g-MA compatibilizer). We believe the differences in mechanical properties are attributed to molecular weight difference of PP used in PP-g-MA reactions. Our PP-g-MA compatibilizer has higher acid number than Polybond. Thus, our PP-g-MA compatibilizer should be able to bond with PA6 with greater efficiency. However, the PP segment in our PP-g-MA compatibilizer has lower molecular weight compared to PP segment in Polybond. As a result, it has less entanglement with PP matrix and lower degree of bonding [4].

Conclusions

- In the PP/MA/DCP reactive blends, acid numbers increase with increasing DCP contents indicating that DCP can drive the MA grafting reaction. At 2 wt% and higher, DCP becomes excessive resulting in no further increase in acid number.
- For PP/PA6/Compatibilizer blends, tensile strength and elongation at break are lower than PA6 but higher than PP due to the effect of PA6 composite reinforcement in the matrix of PP. For impact strength, the values of impact strength of PP/PA6/Compatibilizer blends are clearly higher than PP due to better energy absorption in PA6 phase and its interface.
- The failure mode of these systems in impact tests is cohesive failure on PP matrix. The good adhesion between PA6 and PP is caused by maleated PP compatibilizers.
- PP/PA6/Polybond has a slightly better performance compared to our PP/PA6/PP-g-MA in term of strength, elongation at break, and impact strength. The PP segment in our PP-g-MA compatibilizer has lower molecular weight compared to PP segment in Polybond. As a result, our compatibilizer has less entanglement with PP matrix and lower degree of bonding.

Acknowledgments

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